

Scenario 2: Technical API Architecture

System Architecture Details:

The I.N.A.Y.A.T. system utilizes a tripartite architecture to manage data flow:

- 1. LlamaIndex PropertyGraphIndex: Manages text chunking, embedding generation, and query retrieval.
- 2. Neo4j AuraDB: Stores entity nodes, properties, and relationship links (Cypher query language).
- 3. Mem0 Cloud Service: Provides persistent memory storing personal facts about users using filters={'user_id': user_id}.

Circuit Breaker Details:

A custom CircuitBreaker class (core/resilience.py) is implemented to prevent complete system failure. It acts as a gateway for Mem0 and Neo4j connections. When connection failures exceed a threshold of 3, the circuit breaker state transitions to OPEN, blocking subsequent network requests for 60 seconds and immediately triggering the fallback pipeline (RAG -> direct LLM -> static apologetic string).

API Specifications:

- Model Endpoint: <https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent>
- Embedding Endpoint: <https://generativelanguage.googleapis.com/v1beta/models/gemini-embedding-001:embedContent>
- Authorization Headers: Custom key passed via 'x-goog-api-key' header for Gemini requests.